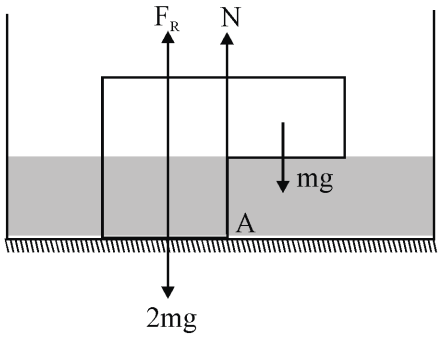
SOLUTION
SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. Ans. (A)

Sol.



FBD when the material just topples,

Balancing torque, about A : $\left(-F_B+2mg\right)\frac{a}{2}=mg.\frac{a}{2}\Rightarrow F_B=mg\Rightarrow \rho_S=\rho_W$

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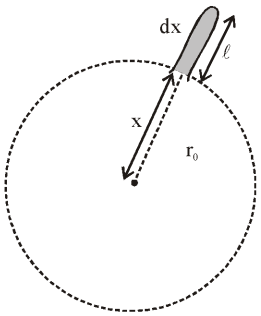
If $\rho_S < \rho_W$, substance will topple clockwise.

If $\rho_S > \rho_W$, substance will not topple & normal will shift to balance the torque by F_B

2. Ans. (A)

Sol. For the section of length dx,

$dT = \rho A\, dx\, \omega^2 x$



$$\therefore \text{ T at centre } = \int_{r_0}^{\left(\frac{\ell}{2}+r_0\right)} \rho A\omega^2 x\,dx = \frac{1}{2}\rho A\omega^2\left[\left(\frac{\ell}{2}+r_0\right)^2 - r_0^2\right]$$

$$= \frac{1}{2}\rho A\omega^2\left[\frac{\ell^2}{4}+r_0^2+\ell r_0 - r_0^2\right] = \frac{1}{2}\rho A\omega^2\left[\frac{\ell^2+4\ell r_0}{4}\right] = \frac{1}{8}\rho A\omega^2\ell\left[\ell+4r_0\right]$$

$$\therefore \text{ Pressure } = \frac{T}{A} = \frac{1}{8}\rho\omega^2\ell\left[\ell+4r_0\right]$$

3. Ans. (A)

Sol. (i) $V_{ice} > V_{water}$ so upper layer falls

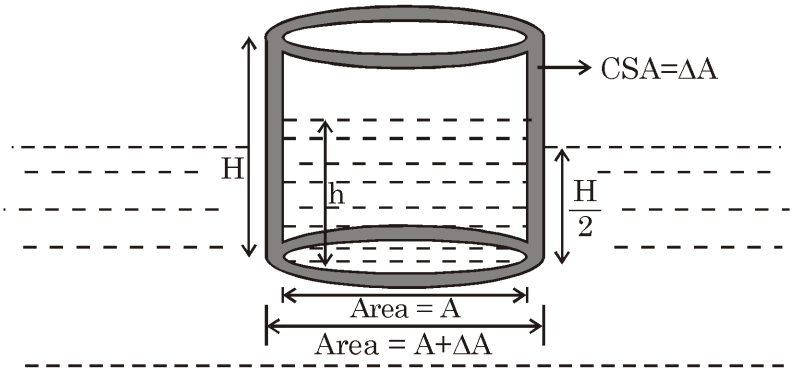
(ii) Displaced volume of water $= \left[M_{ice} - \frac{F_B}{g}\right]\frac{1}{\rho_{water}}$

ice melts then B decreases then level of water rises.

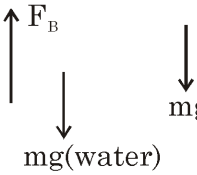
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4. Ans. (A)

Sol.



For equilibrium $\Rightarrow$ FBD $\Rightarrow$



mg (container)

$$F_B = mg_{(Water)} + mg_{(container)}$$

$$\rho_W (A + \Delta A)\frac{H}{2}g = \rho_W hAg + \rho_C \rho_W H\Delta Ag$$

$$hA = \left(\frac{AH}{2}\right) + \left(\frac{\Delta AH}{2}\right) - (\rho_C \Delta AH)$$

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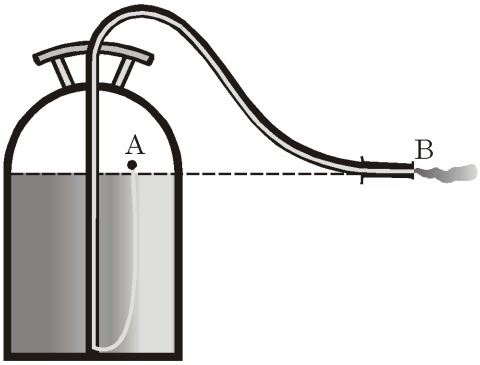
$$h = \frac{H}{2} + \frac{H\Delta A}{A}(0.5 - \rho_C)$$

This shows that $h > \frac{H}{2}$ if $\rho_C < 0.5$

So Ans. (A)

5. Ans. (B)

Sol.



Writing Bernoulli's equation along flow line at point A & B

$$P_{tank} + P_0 + 0 + \frac{1}{2}\rho V_A^2 = P_0 + 0 + \frac{1}{2}\rho V_B^2$$

$$\Rightarrow V_B = \sqrt{\frac{2P_{tank}}{\rho}}$$

[$V_A << V_B$ as area of nozzle is very less as compared to area of tank]

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6. Ans. (C)

Sol. As the height decreases, the rate of flow with which the water is coming out decreases.

7. Ans. (C)

Sol. Pressure at point inside tap is less than P_0

8. Ans. (B)

Sol. $F = \frac{\Delta p}{\Delta t} = \frac{m(v_1 + v_2)}{\Delta t} = V\rho(v_1 + v_2)$ where $\frac{m}{\Delta t} = V$ = volume/second.

9. Ans. (A)

Sol. Let y be the height of liquid at some instant.

Then $-\frac{dy}{dt}$ = constant (given)

From equation of continuity,

$(\pi x^2)\left(-\frac{dy}{dt}\right) = a\sqrt{2gy}$ (a = area of hole)

Here, $\pi\left(\frac{-dy}{dt}\right)$, a and g are constants

Hence, squaring the equation, we get $y = kx^4$.

Multiple Correct Answer Type **4 Q. [4 M (–1)]**

10. Ans. (B, C)

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Sol. $F_H = (\rho gR)\pi R^2 = \rho g\pi R^3 = \frac{3Mg}{2}$

$Fv = (M + m)g$

$F_{net} = \sqrt{[(M + m)g]^2 + \left(\frac{3Mg}{2}\right)^2}$

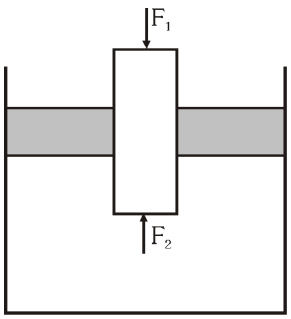
$F_{net} = Mg\sqrt{\frac{9}{4}\left(1 + \frac{m}{M}\right)^2}$

11. Ans. (B,C,D)

Sol. $P_{new} = \rho(g + \frac{g}{2})\,h$ & $B_{new} = P_{new} \times V$

12. Ans. (A,D)

Sol.



$F_2 - F_1$ = upthrust

$\therefore F_2 >$ upthrust

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13. Ans. (A,C)

Sol. $\rho V_A g = (m_{ice} + m_A) g$ and $\rho V_B g = (m_{ice} + m_B) g \Rightarrow \frac{V_A}{V_B} = \frac{m_{ice} + m_A}{m_{ice} + m_B} = \frac{0.9 \times 980 + 4.9 \times 20}{0.9 \times 980 + 1.9 \times 20} = \frac{49}{46}$

Also A will start sinking first.

Linked Comprehension Type (1 Para × 2Q, 1 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

14. Ans. (D)

Sol. Force exerted by liquid = (mass of liquid displaced) × $\sqrt{g^2 + (\omega^2 r)^2}$

$= \rho_w v \times \sqrt{g^2 + (\omega^2 r)^2}$

15. Ans. (B)

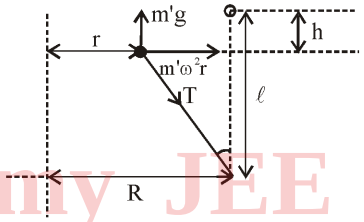
Sol. $R - r = \ell \sin\theta$ ..... (i)

$\ell - h = \ell \cos\theta$ ..... (ii)

$m' = (\rho_w - \rho_{body})v$ & $m'g = T\cos\theta$ ..... (iii)

$m'\omega^2 r = T \sin\theta$ ..... (iv)

from (i), (ii), (iii) & (iv)



$$\frac{\omega^2 r}{g} = \tan\theta = \frac{R - r}{\ell - h}$$

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16. Ans. (C)

Sol. Net force exerted by liquid on styrofoam is buoyant force = $\rho_W Vg = \rho_W(A)\, Lg = \rho_w g \frac{\ell^2 L}{2}$

17. Ans. (D)

Sol. Average pressure on slant surface $P_{avg} = \frac{(P_0 + \rho_w g h_0) + \left(P_0 + \rho_w g h_0 - \frac{l}{\sqrt{2}}\right)}{2} = \left(P_0 + \rho_w g h_0 - \frac{l}{2\sqrt{2}}\right)$

Force on any one of the slant face = $\left[P_0 + \rho_w g(h_0 - \frac{\ell}{2\sqrt{2}})\right] L\ell$

18. Ans. (B)

Sol. Balancing force in vertical

$(M_{styrofoam} + m)g + F_{surface\ Tension} = F_{bouyancy}$

$$\rho_f \left(\frac{1}{2}\ell^2 L\right) g + mg + T\left(2L\sin 45° + \frac{2\ell}{\sqrt{2}}\right) = \rho_W \frac{1}{2}\left(\frac{\ell}{\sqrt{2}} + \sqrt{2}\ell\right)\left(\frac{\ell}{\sqrt{2}} - \frac{\ell}{2\sqrt{2}}\right) Lg$$

$$mg = \frac{-\rho_f g L \ell^2}{2} + \frac{3}{8}\rho_w g L \ell^2 - \sqrt{2}T[L + \ell]$$

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SECTION-II

Numerical Answer Type Question **6 Q. [3(0)]**
(upto second decimal place)

19. Ans. 0.1

Sol. $\rho_1 g h_1 = \rho_2 g h_2$

$$\frac{h_2}{h_1} = \frac{\rho_1}{\rho_2} = \frac{\rho_1}{\rho_1(1-\gamma\Delta T)}$$

$Y = 10^{-5}/°C$, $\Delta T = 100°C$, $h_1 = 300$ cm

$h_2 - h_1 = 0.1$ cm ans.

20. Ans. 4.5

Sol. $P_\ell V_\ell = P_f V_f$

$$P_0(V) = P_f\left(\frac{2V}{3}\right)$$

$$P_f = \frac{3}{2}P_0 = P_0 + \rho_w g h$$

$$\frac{P_0}{2} = \rho_w g h$$

$$\frac{\rho_w g \times 9}{2} = \rho_w g h$$

[h = 4.5 m]

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21. Ans. 100

Sol. $T_{max} = 7 \times 10^6 \times 10^{-6} N = 7N$

$\therefore F_{B_{min}} = (15-7) = 8N$

$\therefore$ Water level its be lowered = 2 cm

$\therefore$ Area occupied by water level at the top = 100 cm^2

$\therefore$ $2t = 100 \times 2 \Rightarrow t = 100$ sec

22. Ans. 110

Sol. $I\alpha = (V\rho g)\frac{\ell}{4}$

where $V = \pi r^2 \ell$

so $\alpha = \frac{\pi r^2 \rho g \ell^2}{4I} = 35\pi = 110$ rad/sec^2

23. Ans. 2500

Sol. $A_1V_1 = A_2V_2$

$10 \times 1 = 5\ V_2$

$V_2 = 2$ m/s

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$$

$$4000 + \frac{1}{2} \times 10^3 (1)^2 = P_2 + \frac{1}{2} \times 1000 (2)^2$$

$P_2 = 2500$

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24. Ans. 40

Sol. $mg = \rho_w \frac{v}{3} g = \rho_A \frac{v}{2} g$ ...(i)

$$\rho_{mixture} = \frac{\rho_w v_0 + \rho_A v_0}{2v_0} = \frac{\rho_w + \rho_A}{2}$$

$$mg = \left[\frac{\rho_w + \rho_A}{2}\right] \times \left[\frac{v}{n}\right] g \text{ ...(ii)}$$

From equation (i) & (ii) $\rho_w \frac{v}{3} g = \frac{vg}{2n}\left[\rho_w + \frac{2\rho_w}{3}\right]; \frac{1}{3} = \frac{1}{2n}\left[\frac{5}{3}\right] \Rightarrow n = \frac{5}{2}$

SECTION-III

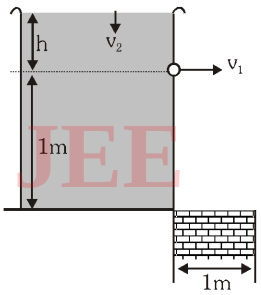
Numerical Grid Type (Ranging from 0 to 9) 3 Q. [4 M (0)]

25. Ans. 5

Sol For minimum velocity at orifice

$$\Rightarrow (v_{min}) \times \sqrt{\frac{2h}{g}} = 1 \Rightarrow v_{min} = \sqrt{5} \text{ m/s}$$

This will give us minimum height for this velocity



$$\Rightarrow \sqrt{2gh_{min}} = \sqrt{5} \Rightarrow h_{min} = \frac{1}{4} m = 0.25m$$

$v_1 = \sqrt{2gh}$ & $v_2 = -dh/dt$

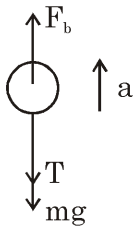
By equation of continuity $av_1 = Av_2$

$$\Rightarrow a\sqrt{2gh} = A\left(-\frac{dh}{dt}\right) \Rightarrow \int_{0.81}^{0.25} \frac{dh}{\sqrt{h}} = \int_0^t \frac{\left(a\sqrt{2g}\right)}{A} dt$$

$\Rightarrow t = 125 \text{ sec} = (5)^3 = (\alpha)^3 \Rightarrow \alpha = 5$

26. Ans. 3

Sol.



$F_b - T - mg = ma$

$F_b - mg - ma = T$

$4 \times 10^{-3} \times 10^3 \times (10 + 2) - 2 \times 10 - 2 \times 2 = T$

$T = 24$

$$V = \frac{m}{\rho} = \frac{2}{0.5 \times 10^3}$$

$V = 4 \times 10^{-3}$

27. Ans. 2

Sol. Equation of continuity $\Rightarrow v_A + v_B = v_C + v_D$

$\Rightarrow v_D = 2m/s$

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SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

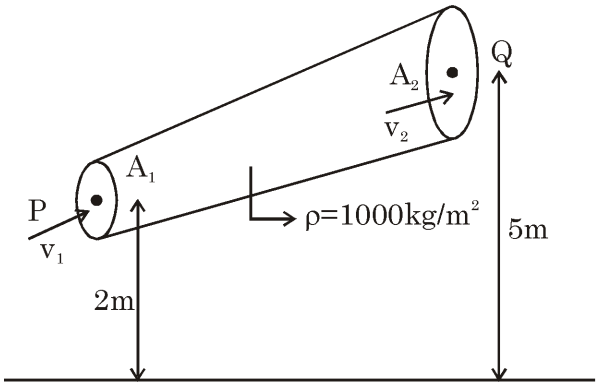
28. Ans. (A) PQ (B) PQ (C) PRST (D) PRST

Sol. If $dgh_1 > P \Rightarrow$ height of liquid will not cross h_1 $dgh_1 < P \Rightarrow$ height of liquid will cross h_1

Subjective Type **2 Q. [4 M (0)]**

29. Ans. + 29625 J/m³, – 30000 J/m³

Sol.



$A_1 = 4 \times 10^{-3}\ m^2$

$A_2 = 8 \times 10^{-3}\ m^2$

$v_1 = 1$ m/sec

$\rho_W = 10^3$ kg/m³

Continuity equation at P & Q

$A_1v_1 = A_2v_2$

$(4 \times 10^{-3})\ (1) = (8 \times 10^{-3})\ (v_2)$

$$v_2 = \frac{1}{2}\text{ m/sec}$$

Work done per unit volume by pressure = ΔP = pressure difference

Applying Bernoulli's theorem between P & Q

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

$$P_1 - P_2 = \rho g\left(h_2 - h_1\right) + \frac{\rho}{2}\left(v_2^2 - v_1^2\right) = 10^4\left(5-2\right) + 500\left(\frac{1}{4} - 1\right)$$

$$= 30000 - \frac{1500}{4} = 29625\ \text{J/m}^3$$

Work done per unit volume by gravity force $= \frac{mgh}{v} = (\rho gh)$

So, work done/volume = $\rho g\ (h_1 - h_2)$

$= 1000 \times 10\ [2 - 5]$

$= -30000$ J/m³

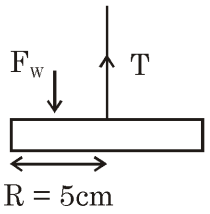
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30. **Ans.** $h_1 = \frac{2(3+\pi)}{15\pi} = 0.26$; $h_2 = \frac{3+\pi}{10\pi} = 0.195$

Sol. Plug opens when level of water reaches 'h'.

- FBD of plug



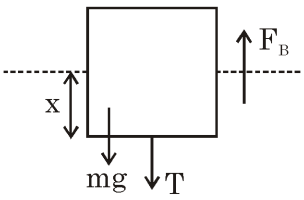
$F_W = T$

$P.\pi R^2 = T$

$$T = \rho_W ghA = 1000 \times 10 \times h \times \pi\left(\frac{5}{100}\right)^2$$

$T = 25\ \pi h$

- FBD of float



$F_B = mg + T$

$\rho_W gxA = 3 \times 10 + T$

$$10000x\pi \times \left(\frac{10}{100}\right)^2 = 30 + 25\ \pi h$$

$100\ \pi x = 30 + 25\ \pi h$

$100\pi\ (h - 0.1) = 30 + 25\ \pi h$

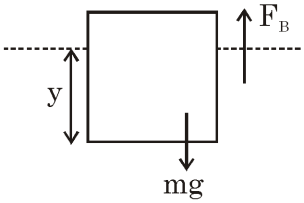
$75\ \pi h = 30 + 10\ \pi$

$$h = \frac{(3+\pi)10}{75\pi} = \left(\frac{2(3+\pi)}{15\pi}\right) m$$

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- To again close the plug

$mg = F_B$ (for the float) $\rightarrow$ [When water has risen to height y for the float]



$F_B = mg$

$\rho_W gyA = 3g$

$$1000 \times 10 \times y \times \pi \times \left(\frac{10}{100}\right)^2 = 3g$$

$100\pi y = 3g$

$$y = \frac{30}{100\pi} = \frac{3}{10\pi} m$$

$$\text{So } h = 0.1\ m + \frac{3}{10\pi} m$$

$$= \left(\frac{1}{10} + \frac{3}{10\pi}\right) m$$

$$h = \left(\frac{\pi+3}{10\pi}\right) m$$

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